

DISASTER -GUARD

AI

NEURAL-NEXUS

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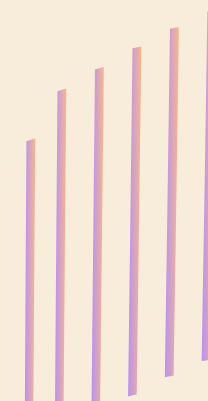
Problem Statement



- **Problem Statement**

Using the Disaster images Dataset ,design and develop an advance AI-Driven system to classify the disaster type and extracting the meaningful visual patterns and reflect that reflect severity and contextual damage .

Disaster classification is the systematic categorization of catastrophic events based on their origin, nature, speed of onset, and impact. A disaster is defined as a serious disruption to a community's functioning that exceeds its ability to cope using its own resources, often resulting in widespread human, material, economic, or environmental losses.





Problem Solution:

The technical challenge addressed by DisasterGuard AI centers on feature warping and geometric distortion resulting from non-square webcam telemetry (16:9) being force-resized into fixed 150x150 convolutional kernels, which induces a systematic misclassification bias, particularly horizontal artifact skewing toward Water Disaster. Our solution implements a high-concurrency FastAPI inference pipeline that utilizes a dynamic center-cropping algorithm to isolate square ROIs (Regions of Interest) before passing the float32-normalized pixel arrays into a Sequential CNN engine. This architecture produces a 6-node Softmax probability distribution, which is processed through an asynchronous feedback API to log prediction telemetry and cross-verify actual vs. predicted labels, ensuring high-fidelity situational awareness while eliminating feature distortion at the preprocessing layer



Technical Approach

Technologies To be Used

Python
Pandas & NumPy
Scikit-learn
TensorFlow / Keras
Matplotlib
PIL (Image Processing)

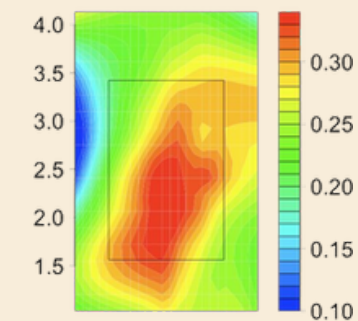
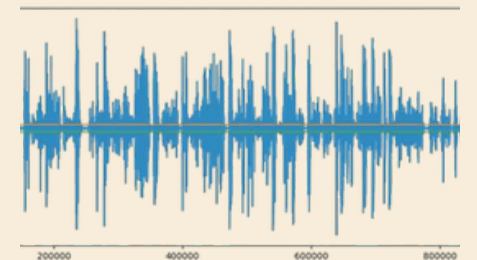
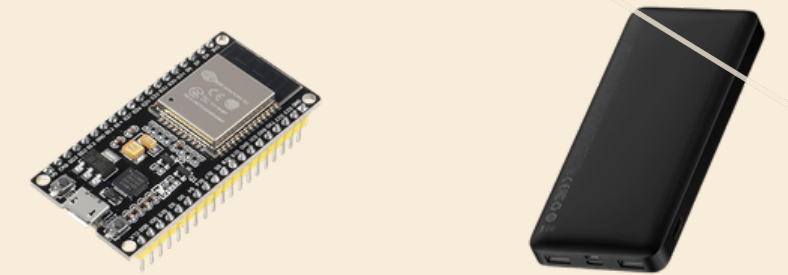
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Text Classification
TF-IDF Vectorization
Logistic Regression model
Evaluation metrics:
Accuracy
Precision
Recall
F1 Score

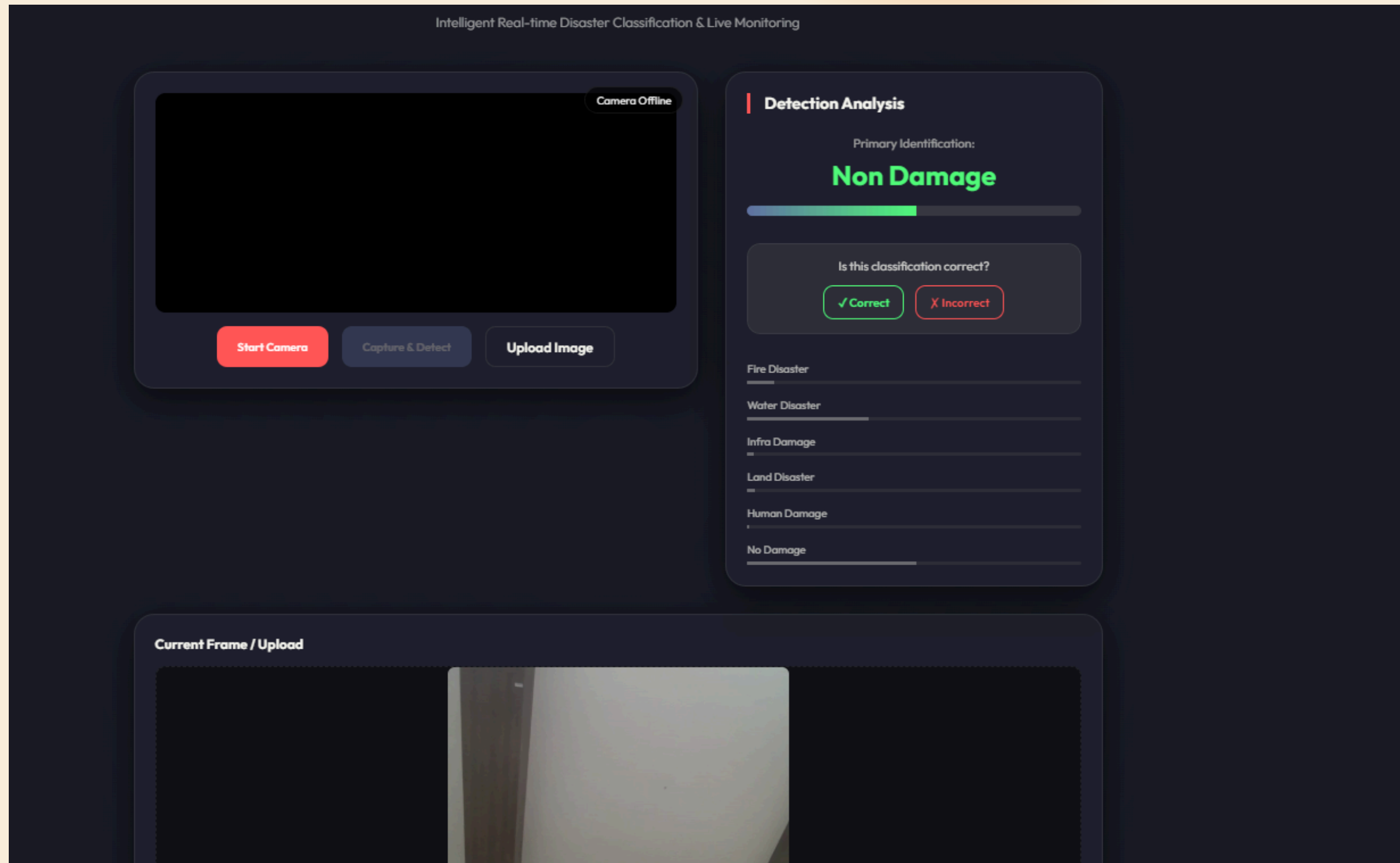
02

Image Classification
Image preprocessing with ImageDataGenerator
Transfer learning using VGG19
Fine-tuning neural network layers
Callbacks for training optimization:
EarlyStopping
ReduceLROnPlateau
ModelCheckpoint
TensorBoard logging

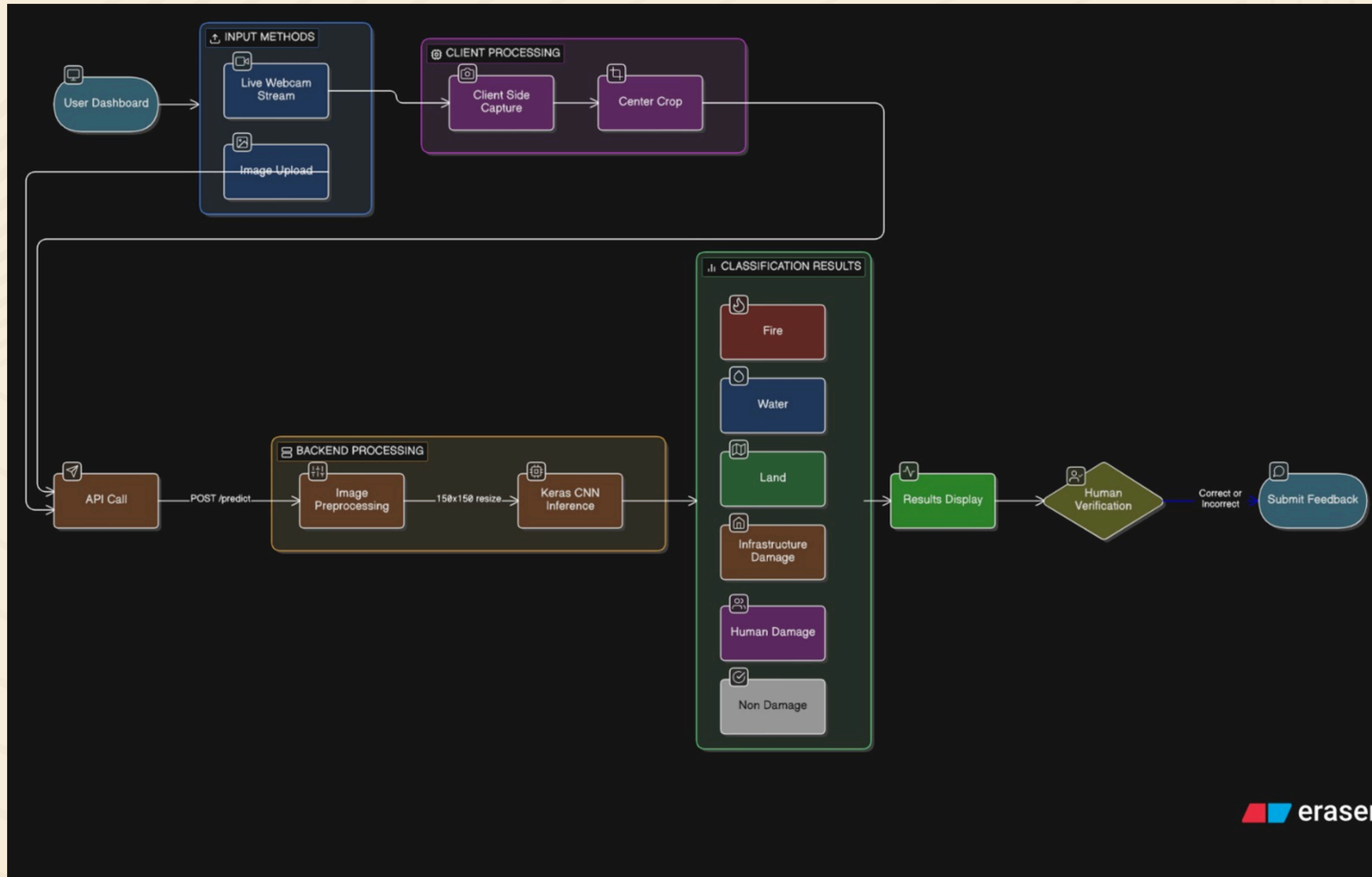
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Prototype



Integrated Workflow

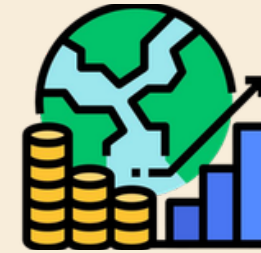


Impacts and Benefits



Social Impact

- Improves disaster response by rapid detection of survivors trapped under debris during earthquakes , building collapses, or explosions.
- Provides rescuers with real time location insights, helping them locate victims faster and increasing survival chances.



Economic Impact

- Offers a low cost and scalable alternative to expensive radar-based life detection systems used in search and rescue operations.
- Reduces operational costs and improves efficiency of the rescue missions.
- Supports deployment for government disaster management agencies and humanitarian organizations.



Target Audience

- Disaster response agencies(NDRF/SDRF)
- Search and Rescue Teams
- Military and Emergency Response Units
- Disaster Management Authorities
- Humanitarian and Relief Organizations.

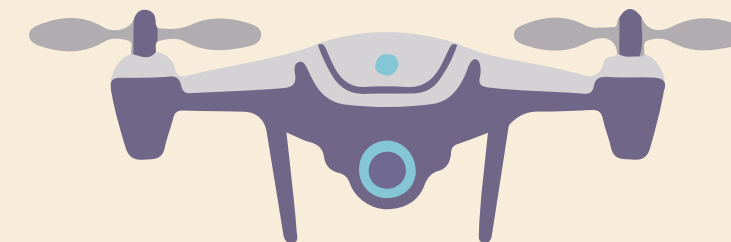


Environmental Impact

- Reducing unnecessary excavation and damage to surrounding structures by targeting high probability survivor zones.
- Optimizes resource usage during rescue , reducing energy and equipment waste.
- Encourages efficient and sustainable disaster management practices.

Future Scope

- **Real-time Hybrid Classification:** Advanced deep learning models will integrate live drone imagery, social media, and satellite data to classify disasters (e.g., floods vs. earthquakes) within seconds to trigger automated response protocols.
- **Through-Wall RF Human Detection:** Future rescue drones will use AI-driven RF-pose sensing to detect the precise location and vital signs of victims trapped under rubble, even in zero-visibility conditions, without needing line-of-sight.
- **Autonomous Swarm Robotics:** Multiple small drones/robots equipped with RF sensors will work in coordination to map damaged areas and identify human survivors in areas too hazardous for human rescuers.
- **Adaptive Structural Damage Assessment:** AI systems will analyze pre- and post-disaster data to differentiate between damaged, collapsed, and intact structures, allowing rescue teams to prioritize high-probability survival zones.
- **Predictive Rescue Planning:** Combining classification models with AI will enable predicting the shortest rescue paths and potential secondary dangers, significantly lowering casualties during the critical first hours of a disaster.



Research and References

ResNet-50 (Residual Network): Frequently cited for high performance in image classification, with enhanced variants (like improved ResNet-50) used for multi-disaster types such as earthquake, fire, flood, and landslide.

EfficientNet: Used for classification of natural disaster imagery due to its compound scaling approach, which provides high efficiency in identifying floods, earthquakes, cyclones, and wildfires.

Inception V3, VGG16, VGG19: Evaluated in studies as highly effective for classifying optical satellite images into disaster events, with Inception V3 often performing best in benchmarking, sometimes achieving 92% accuracy.

- **Links:**

- **WiFiSensing for Human Activity Recognition (Research Paper)**
<https://arxiv.org/abs/1608.07546>
- **RF-Capture: Through-Wall Human Detection (MIT Research)**
<https://people.csail.mit.edu/fadel/papers/rfcapture-sigcomm15.pdf>
- **ESP32 CSI Tool (RF Signal Research)**
<https://github.com/StevenMHernandez/ESP32-CSI-Tool>
- **WiFi Sensing Research Paper**
<https://dl.acm.org/doi/10.1145/2619239.2630560>
- **RF Human Detection Research**
<https://ieeexplore.ieee.org/document/7470937>